EE 156: FUNDAMENTALS OF ELECTRICAL CONTROLS AND PLC SYSTEMS

Chapter 6: Developing a Wiring Diagram

6.1 Introduction

What is a Wiring Diagram?

- A **visual representation** of components and wires related to an electrical connection
- Shows the **physical layout** of components
- Shows **wire connections** between components

Purpose of Wiring Diagrams

Purpose	Description
Communication	Facilitates communication between electrical engineers
Installation	Guides the physical installation of equipment
Repairs	Helps in making repairs and modifications
Verification	Confirms proper design and implementation
Safety	Ensures compliance with safety regulations

Difference from Schematic

Feature	Schematic	Wiring Diagram
Shows	Electrical sequence	Physical layout
Focus	Circuit function	Component connection
Reading	Logical flow	Physical routing

6.2 Development Process

Step-by-Step Approach

Step	Action
1	Understand the schematic diagram
2	Identify all components
3	Visualize the physical panel layout
4	Draw components in their physical positions
5	Connect components using a numbering system
6	Label all wires and terminals

Visualization Technique

- Use **imagination** to visualize actual relays and contacts
- Imagine them **mounted in a panel**
- Picture wires **connecting the various components**

6.3 Numbering System

Purpose of Numbering

- Used to identify **connections** between components
- Ensures wires are connected to the **correct terminals**
- Makes troubleshooting **easier**

Numbering Rules

Rule	Description	Example
1	Each time a component is crossed, the number must change	Wire #1 -> Switch -> Wire #2
2	All connected components get the same number	All points connected together = same number
3	Never use a number set more than once on the same rung	Unless there is a parallel connection
4	Numbers in wiring diagram match schematic numbers	Same component labeled the same way

Basic Numbering Example

```
L1                                         N
 |                                          |
 |                                          |
 |    +-----+    +-------+    +------------+|
 |    |     |    |       |    |            ||
 +----+  1  +----+  2    +----+   Load     ||
 |    |     |    |       |    |            ||
 |    +--+--+    +---+---+    +------------+|
 |       |           |                      |
 |       |    +------+------+               |
 |       |    |             |               |
 |       +----+  3          |               |
 |            |             |               |
 |            +-------------+               |
 |                                          |
 L1                                         N
```

Numbering Explained:

- Point 1: Between L1 and first component
- Point 2: Between first and second component
- Point 3: Between second component and load

6.4 Developing Wiring Diagrams

General Process

1. **Start with the schematic**
2. **Layout components** as they would appear in panel
3. **Draw wire connections** between components
4. **Assign numbers** to all connection points
5. **Label all terminals** clearly

Example: Alarm Silencing Circuit

Circuit Components:

Component	Function
Pushbuttons	Manual control inputs
Relay (CR)	Control relay for alarm
Contacts	Switching elements
Horn	Audio alarm
Light	Visual alarm indicator
Power Supply	Provides circuit power

Step 1: Labelled Schematic

```
L1                                              L2
 |                                               |
 |                                               |
 |    +-----+    +-------+    +-----------------+|
 |    |     |    |       |    |                 ||
 +----+ PB1 +----+ CR    +----+   Alarm Horn    ||
 |    |(NO) |    |       |    |                 ||
 |    +--+--+    +---+---+    +-----------------+|
 |       |           |                            |
 |       |    +------+------+    +---------------+|
 |       |    |             |    |               ||
 |       +----+ CR NO       +----+ Warning Light ||
 |            |   (Seal)    |    |               ||
 |            |             |    |               ||
 |            +-------------+    +---------------+|
 |                                               |
 |            +-------------+    +---------------+|
 |            |             |    |               ||
 +------------+ PB2         +----+ Horn Circuit  ||
              |   (NC)      |    |               ||
              |             |    |               ||
              +-------------+    +---------------+|
                                                  |
 L1                                              L2
```

Step 2: Component Labelling

Label each component with a unique identifier:

Component	Label	Description
Pushbutton #1	PB1	Normally Open (Start)
Pushbutton #2	PB2	Normally Closed (Stop)
Control Relay	CR	Coil and contacts
Alarm Horn	H	Audio alarm
Warning Light	L	Visual alarm

Step 3: Add Wire Numbers

Apply numbering rules:

```
L1                                              L2
 |                                               |
 |                                               |
 |    +-----+    +-------+    +-----------------+|
 |    |     |    |       |    |                 ||
 +----+ PB1 +----+ CR    +----+   Alarm Horn    ||
 |   |101  |   |102     |   |103               ||
 |    +--+--+    +---+---+    +-----------------+|
 |       |           |                            |
 |       |    +------+------+    +---------------+|
 |       |    |             |    |               ||
 |       +----+ CR NO       +----+ Warning Light ||
 |            |  (Seal)     |    |               ||
 |            |             |    |               ||
 |            +-------------+    +---------------+|
 |                                               |
 |            +-------------+    +---------------+|
 |            |             |    |               ||
 +------------+ PB2         +----+ Horn Circuit  ||
              |  (NC)       |    |               ||
              |             |    |               ||
              +-------------+    +---------------+|
                                                  |
 L1                                              L2
```

Step 4: Create Wiring Diagram

Layout Components in Physical Position:

```
Panel Layout (Top View)

+------------------------------------------------------------+
|                                                              |
|    +---------+             +-------------+               |
|    |         |             |             |               |
|    |  CR     |             |  Alarm Horn |               |
|    |  Relay  |             |             |               |
|    |         |             |             |               |
|    +----+----+             +------+------+               |
|         |                         |                      |
|    +----+----+             +------+------+               |
|    |         |             |             |               |
|    |  PB1    |             |  Warning    |               |
|    |  (NO)   |             |  Light      |               |
|    |         |             |             |               |
|    +----+----+             +------+------+               |
|         |                         |                      |
|    +----+----+                    |                      |
|    |         |                    |                      |
|    |  PB2    |                    |                      |
|    |  (NC)   |                    |                      |
|    |         |                    |                      |
|    +---------+                    |                      |
|                                   |                      |
|    +-------------------------------+---------------+      |
|    |                                               |      |
|    |              Terminal Block                   |      |
|    |    L1  101  102  103   N                      |      |
|    +-----------------------------------------------+      |
|                                                              |
+------------------------------------------------------------+
```

6.5 Wiring Diagram Construction Rules

Rule 1: Component Layout

- Place components in their **actual physical positions**
- Consider **panel space** and **accessibility**
- Group related components together

Rule 2: Wire Routing

- Use **numbered wires** for identification
- Follow **logical paths** between components
- Avoid crossing wires when possible

Rule 3: Terminal Identification

- Label **all terminals** clearly
- Use numbers that **match the schematic**
- Include **wire color** information when needed

Rule 4: Power Distribution

- Show **main power** connections clearly
- Use **bus bars** for common connections
- Indicate **wire sizes** and **types**

6.6 Connection of Components

Terminal Block Connections

```
Terminal Block

+-----+-----+-----+-----+-----+
|     |     |     |     |     |
|  1  |  2  |  3  |  4  |  5  |
|     |     |     |     |     |
+-----+-----+-----+-----+-----+
|     |     |     |     |     |
|  6  |  7  |  8  |  9  |  10 |
|     |     |     |     |     |
+-----+-----+-----+-----+-----+
```

Wiring Between Components

From	Wire #	To	Description
L1	101	PB1 (input)	Power to pushbutton
PB1 (output)	102	CR (coil)	Control signal
CR (coil output)	103	Horn	Alarm signal
Horn	N	N	Return path

6.7 Connecting Control Circuit

Step-by-Step Connection

Step 1: Power Supply Connection

- Connect L1 and N to terminal block
- Use appropriate wire size
- Include overcurrent protection

Step 2: Pushbutton Connection

- Connect L1 to PB1 (NO)
- Connect PB1 output to terminal point

Step 3: Relay Connection

- Connect from PB1 to CR coil
- Connect CR coil output to next point

Step 4: Output Device Connection

- Connect from CR output to load (horn/light)
- Connect load to N return

Step 5: Seal Circuit Connection

- Connect CR NO contact in parallel with PB1
- This maintains circuit after button release

6.8 Practical Tips

Panel Layout Considerations

Factor	Consideration
Accessibility	Components should be easy to access
Heat Dissipation	Allow space for heat dissipation
Wiring Space	Provide room for wire connections
Safety	Maintain clearance for safe operation
Future Maintenance	Allow space for future modifications

Wire Identification

Practice	Reason
Number each wire	Easier troubleshooting
Use color coding	Quick visual identification
Create wire list	Documentation for future reference
Label both ends	Avoid confusion during connections

6.9 Example: Wiring Diagram Development

Circuit: Alarm Silencing Circuit

Components:

1. Power supply (120V AC)
2. Start pushbutton (PB1 - NO)
3. Stop pushbutton (PB2 - NC)
4. Control relay (CR)
5. Alarm horn
6. Warning light

Connection Sequence:

Step	Action	Wire #
1	L1 to PB1 (NO) input	101
2	PB1 output to CR coil	102
3	CR coil output to horn	103
4	Horn to PB2 (NC)	104
5	PB2 output to N	105
6	CR NO contact in parallel with PB1	102-103

Final Wiring Diagram

```
+------------------------------------------------------------+
|                                                             |
|  L1 -------+                    N -----------------------+ |
|            |                                            | |
|  +---------+---------+     +----------------------------+| |
|  |                   |     |                            || |
|  |   Terminal 101    |     |   Terminal 105             || |
|  |                   |     |                            || |
|  +---------+---------+     +--------------+-------------+| |
|            |                              |              |
|  +---------+---------+                     |              |
|  |                   |                     |              |
|  |      PB1          |                     |              |
|  |     (NO)          |                     |              |
|  |                   |                     |              |
|  +---------+---------+                     |              |
|            |                               |              |
|  +---------+---------+     +--------------+-------------+ |
|  |                   |     |                            | |
|  |   Terminal 102    |     |      PB2                   | |
|  |                   |     |     (NC)                   | |
|  +---------+---------+     |                            | |
|            |               +--------------+-------------+ |
|  +---------+---------+                    |               |
|  |                   |                    |               |
|  |    CR Relay       |                    |               |
|  |   (Coil)          |                    |               |
|  |                   |                    |               |
|  +---------+---------+                    |               |
|            |                              |               |
|  +---------+---------+                    |               |
|  |                   |                    |               |
|  |   Terminal 103    |                    |               |
|  |                   |                    |               |
|  +---------+---------+                    |               |
|            |                              |               |
|  +---------+---------+     +--------------+-------------+ |
|  |                   |     |                            | |
|  |   Alarm Horn      |     |   Warning Light            | |
|  |                   |     |                            | |
|  +---------+---------+     +--------------+-------------+ |
|            |                              |               |
|  +---------+---------+     +--------------+-------------+ |
|  |                   |     |                            | |
|  |   Terminal 104    |     |   Terminal 103             | |
|  |                   |     |                            | |
|  +---------+---------+     +--------------+-------------+ |
|            |                              |               |
|            +-------------+----------------+               |
|                          |                                 |
|                  +-------+-------+                         |
|                  |               |                         |
|                  |  Terminal N   |                         |
|                  |               |                         |
|                  +---------------+                         |
|                                                             |
+------------------------------------------------------------+
```

6.10 Checklist for Wiring Diagrams

Item	Check
All components from schematic are included	■
Components are in logical physical positions	■
All wires are numbered correctly	■
Numbers match schematic	■
Connections are clearly shown	■
Power supply connections are correct	■
Return paths are shown	■
Seal circuits are properly wired	■
Safety devices (overloads) are included	■
Clear labels are used	■
Wire sizes are indicated	■
Color coding is specified	■

Key Terminology

Term	Definition
Wiring Diagram	Visual representation of physical connections
Numbering System	Method of identifying wire connections
Terminal Block	Connection point for multiple wires
Panel Layout	Physical arrangement of components
Seal Circuit	Circuit that maintains itself
Wire List	Documentation of all wire connections

Review Questions

1. What is the purpose of a wiring diagram?
2. How does a wiring diagram differ from a schematic diagram?
3. What are the rules for numbering wires in a wiring diagram?
4. Why is it important to use the same numbers as in the schematic?
5. Describe the process of developing a wiring diagram.
6. What should be considered when laying out components in a panel?
7. Why is it important to visualize the physical layout?
8. What is a terminal block and how is it used?
9. How do you connect components using a numbering system?
10. What are the benefits of using a wiring diagram for installation?

END OF CHAPTER 6 NOTES